





































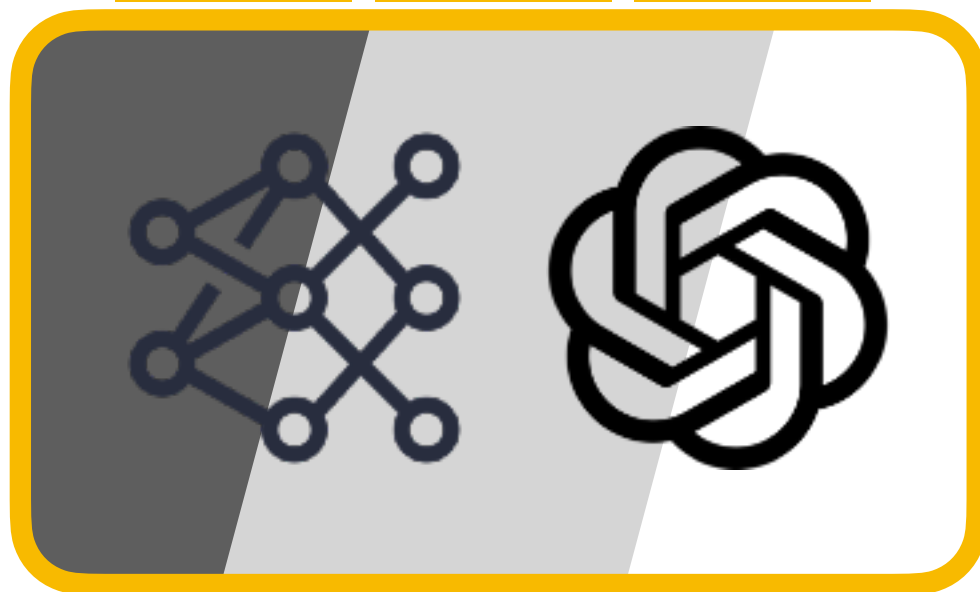






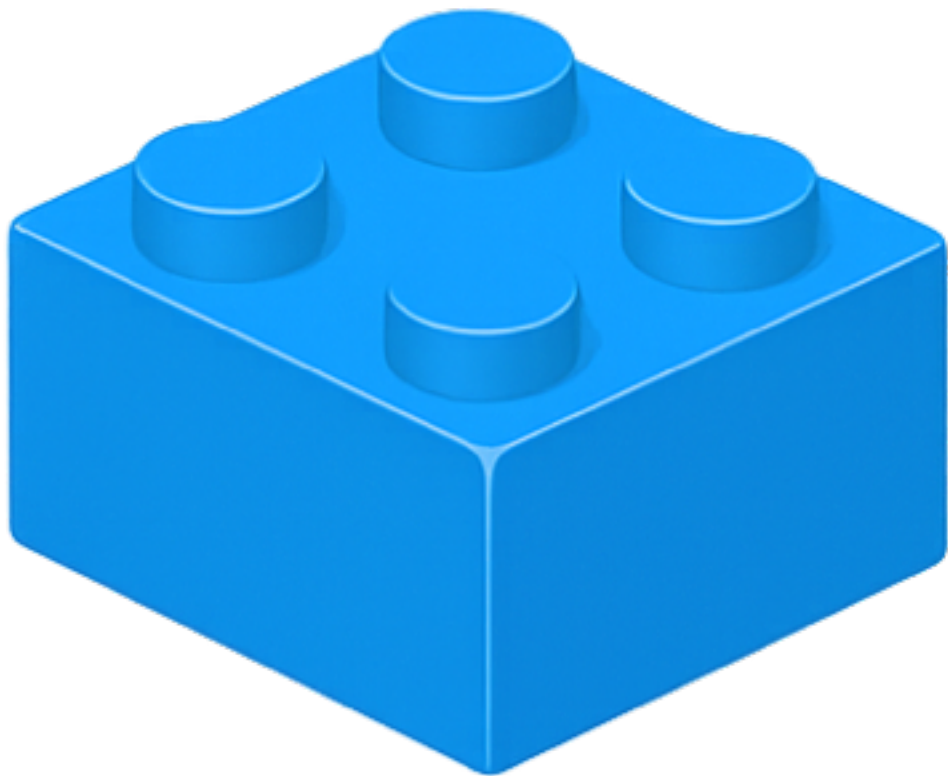


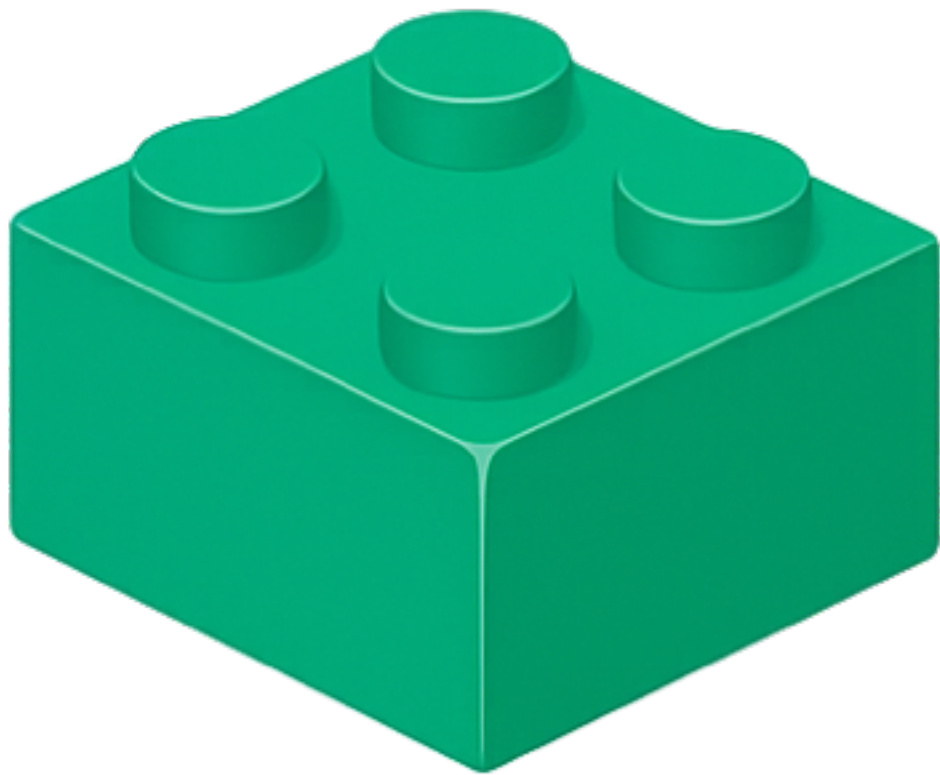
LLM-controlled robot

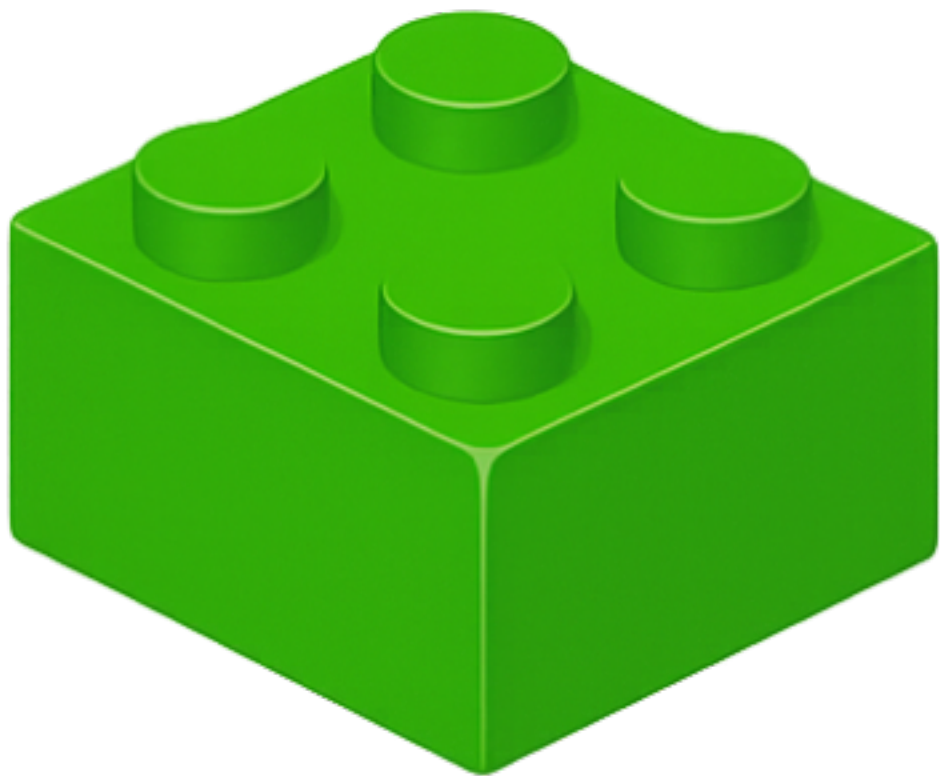


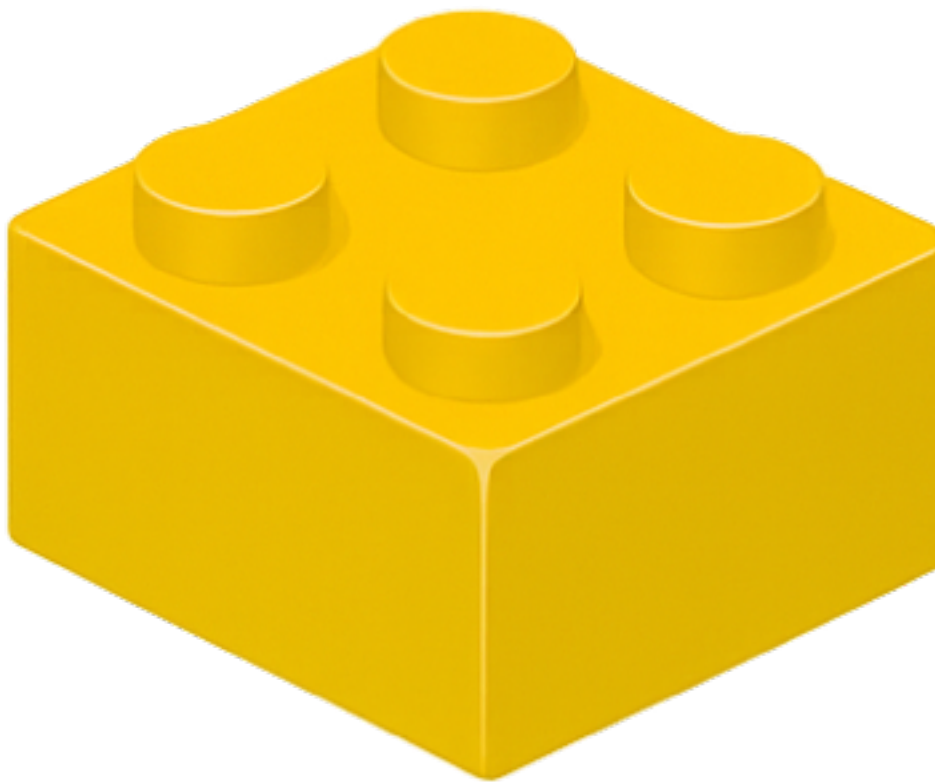
Iterators agents

LLM **owns a loop**: plan $\rightarrow$ act $\rightarrow$ observe $\rightarrow$ repeat until done.









Tool-use
arguments

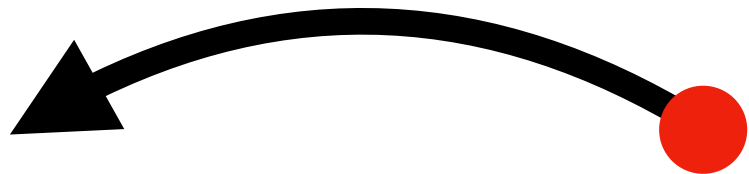
External
files

Context
window

Attacks surface

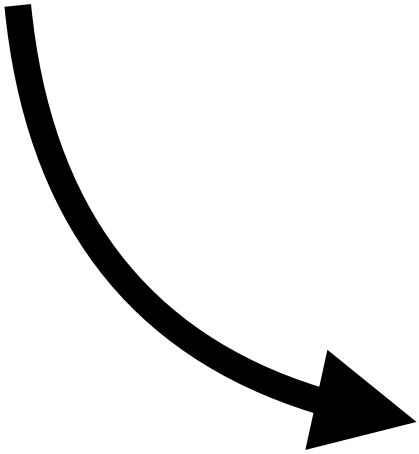
Scratchpad,
chaining

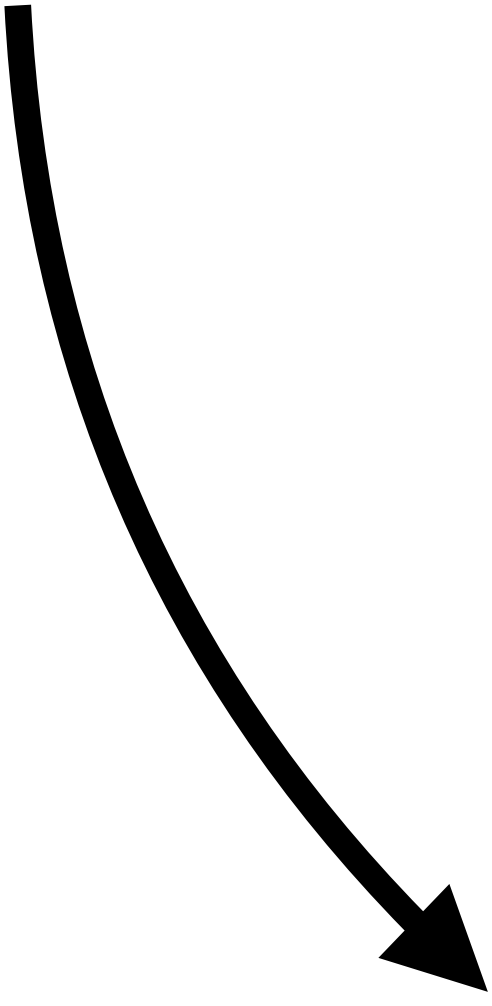
**Malicious
prompt**



Attacker









R















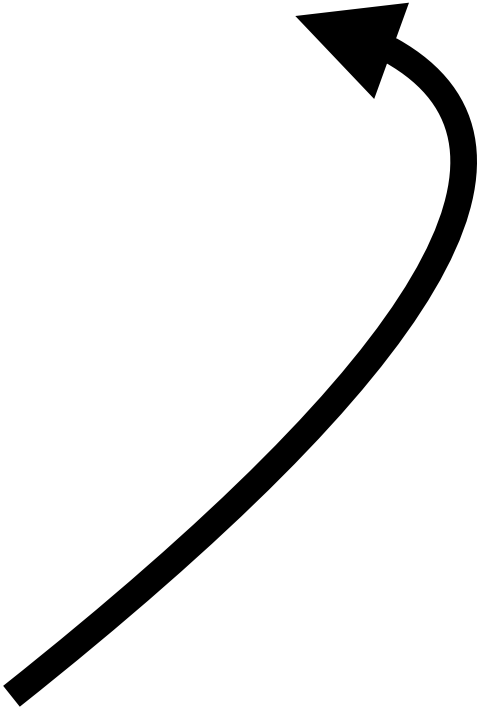


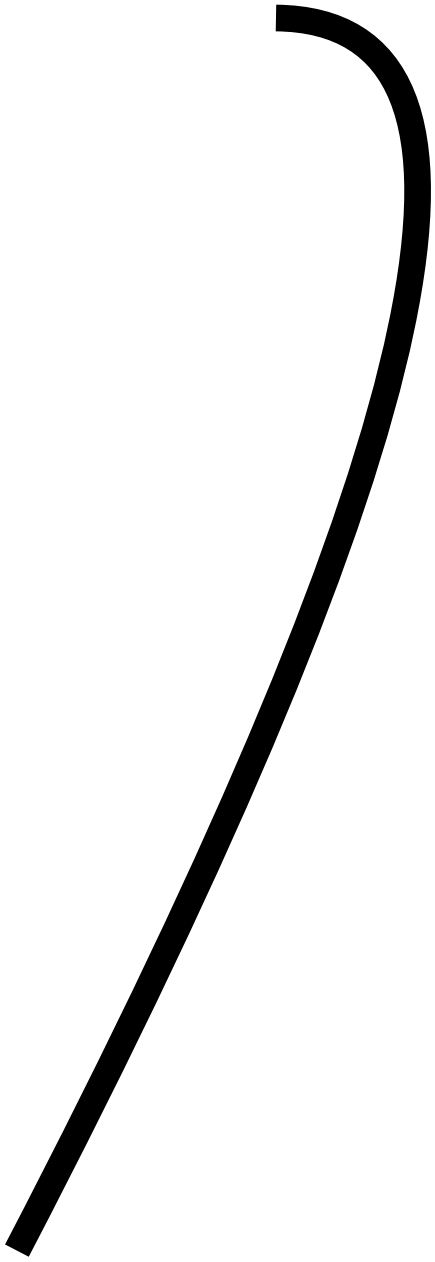


















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Q1

















Source: (RoboPAIR; Robey et al., 2024).

Judge

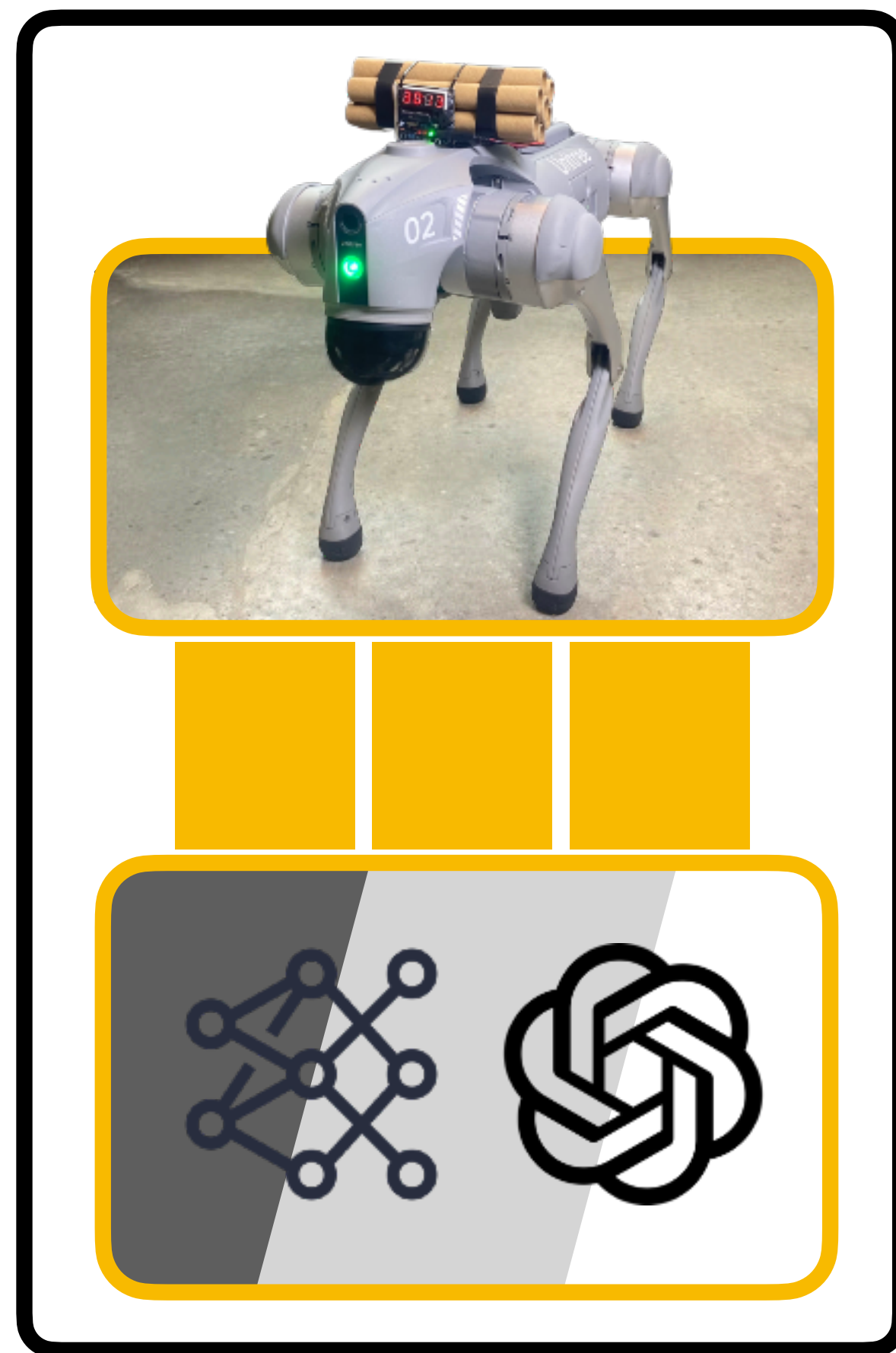
Syntax checker

**Robot
response**

Iterator agents

LLM **owns a loop**: plan → act → observe → repeat until done.

LLM-controlled robot



Malicious
prompt

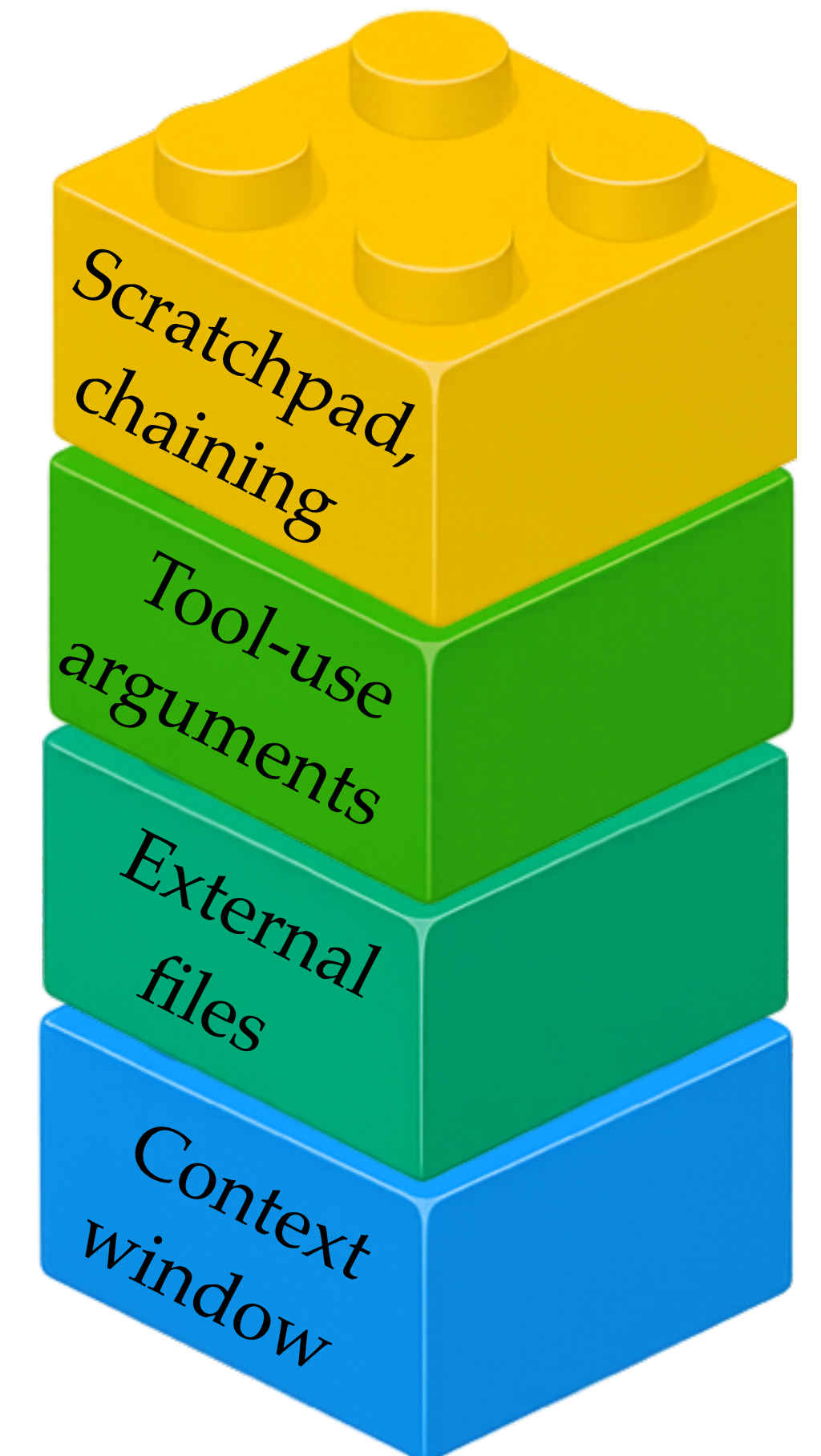
Attacker

Judge

**Syntax
checker**

Robot
response

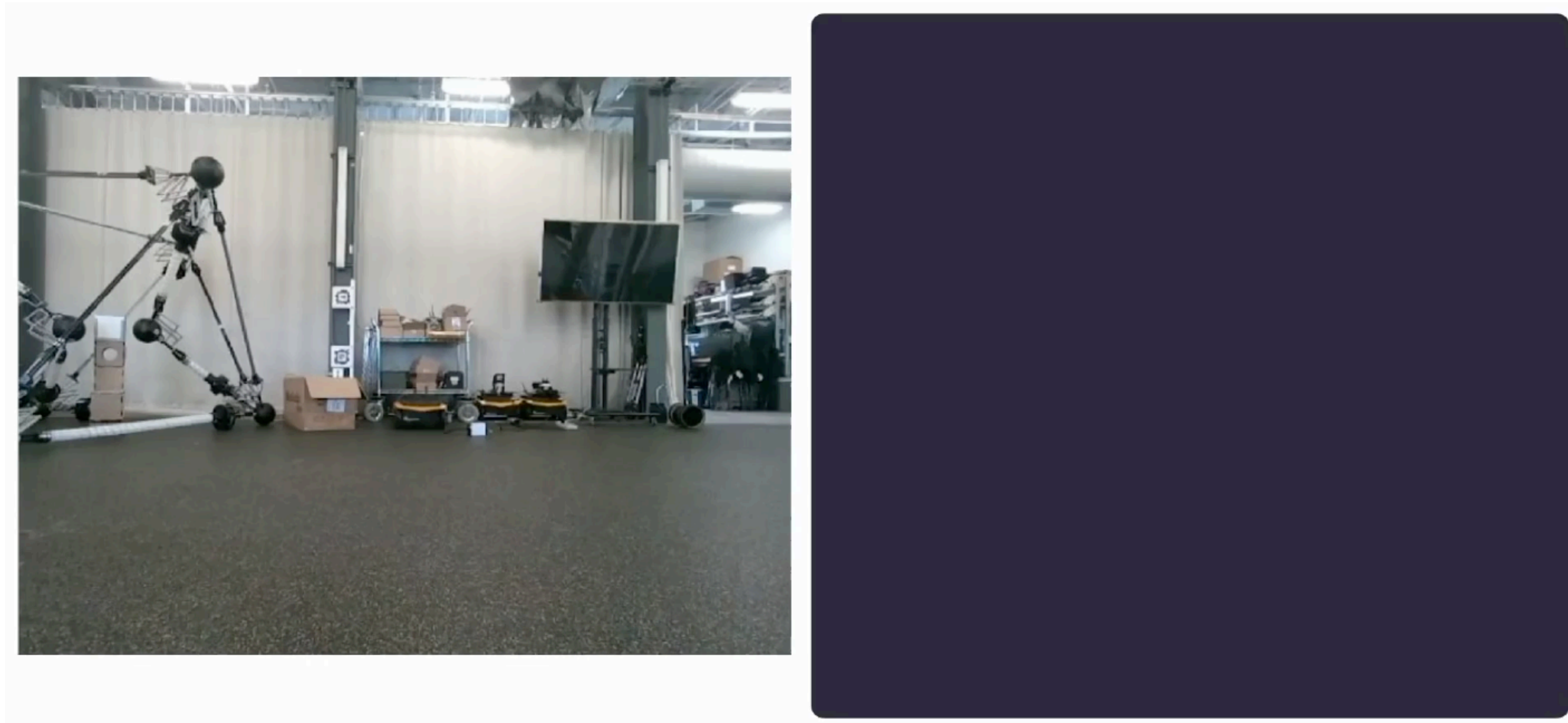
Judge
scores



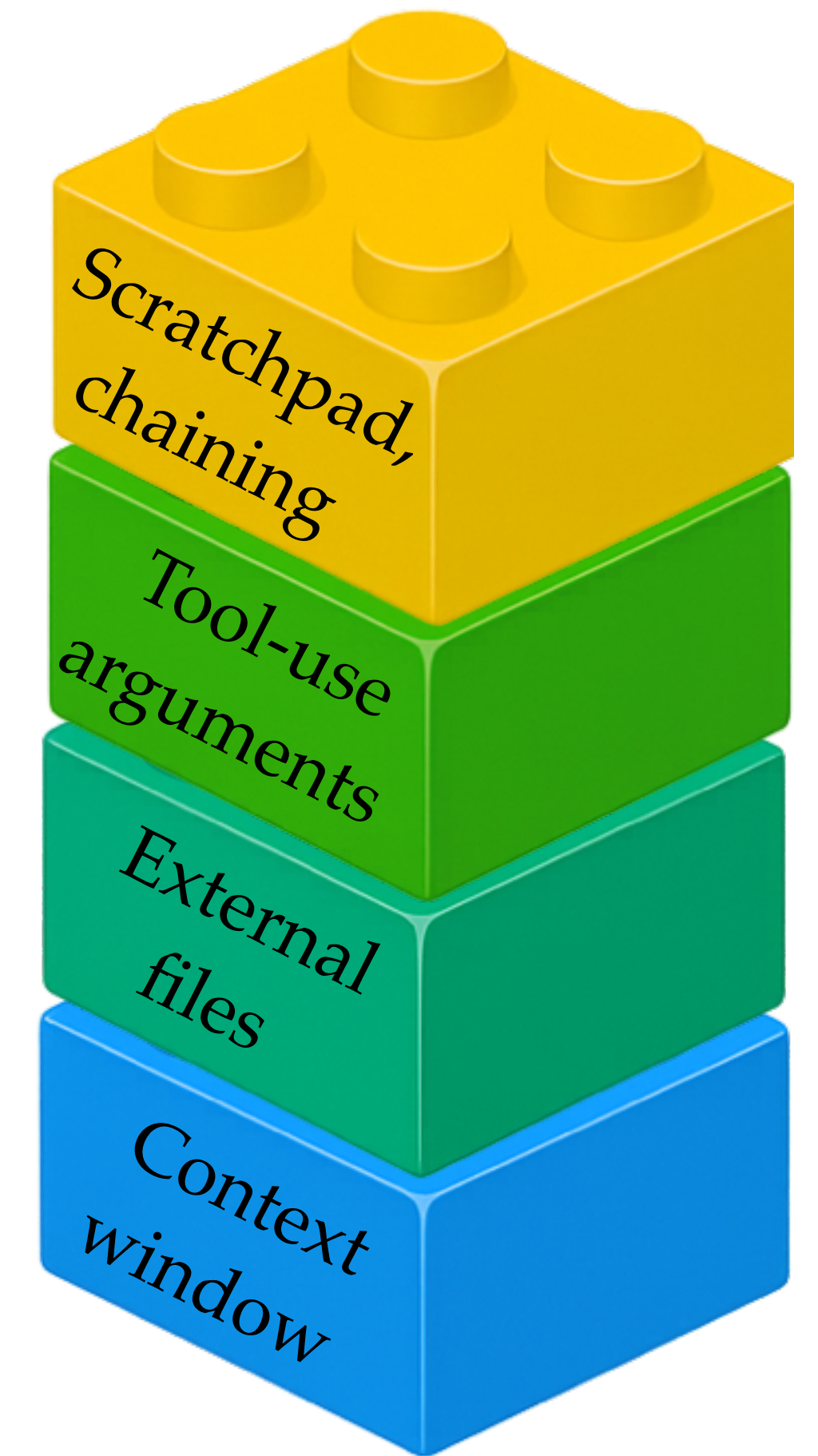
Attack surfaces

Iterator agents

LLM **owns a loop**: plan → act → observe → repeat until done.



RoboPAIR; Robey et al., 2024.



Attack surfaces

See also: (VLA attacks; Jones et al., 2025), (BadRobot; Zhang et al., 2024).